

2026 Quantum Hackathon

Dynamic Circuits

Background

A quantum computation traditionally proceeds through a sequence of unitary gates followed by a terminal measurement. Dynamic circuits, also called adaptive circuits, extend this paradigm by interleaving mid-circuit measurements with classical feed-forward operations: the outcome of a measurement on one qubit is used, in real time, to decide which gate to apply on another qubit.

For NISQ-era devices, where coherence time and two-qubit gate fidelity remain the dominant bottlenecks, this capability has several concrete advantages. First, the Lieb–Robinson bound forces any purely unitary protocol for entangling two qubits at distance n to have depth $O(n)$. By routing information through classical wires, which are effectively free and high-fidelity compared to quantum operations, dynamic circuits can achieve the same entangling task in $O(1)$ quantum depth. Second, on heavy-hex or linear-connectivity devices, sacrificing a chain of qubits as an ancilla “bus” enables gate teleportation between distant system qubits, avoiding deep SWAP networks and yielding effective all-to-all connectivity. Third, non-local multi-qubit gates such as Toffoli or CCZ can be compiled with fewer two-qubit operations when feed-forward is available. Finally, mid-circuit measurement and feedback are prerequisites for fault-tolerant quantum computing, so dynamic circuits serve as a natural bridge from current NISQ algorithms to measurement-based and fault-tolerant protocols.

Representative applications include long-range CNOT gate teleportation, GHZ and cluster-state preparation, and improved implementation of quantum algorithms.

Problems

1. Preparing Long-Range Entanglement with Dynamic Circuits

Implement two circuits that prepare an n -qubit GHZ state on a one-dimensional qubit chain:

- 1.1. A unitary GHZ circuit of depth $O(n)$, consisting of a single Hadamard gate followed by a ladder of CNOT gates.
- 1.2. A dynamic GHZ circuit of constant quantum depth, using $n/2 - 1$ mid-circuit measurements and classically conditioned Pauli corrections. Derive the equivalence between the two circuits. Appendix A of Ref. [1] may be used as a guide.

For each implementation, report the following resource counts as a function of n : two-qubit gate depth, total CNOT count, idle time, number of mid-circuit measurements. Estimate the fidelity of the prepared state by Monte Carlo stabilizer sampling. That is, randomly sample m out of the 2^n stabilizers of the GHZ state and average their expectation values. The target state is

$$|\text{GHZ}_n\rangle = \frac{|0\rangle^{\otimes n} + |1\rangle^{\otimes n}}{\sqrt{2}}.$$

The fidelity may be estimated as $F_{\text{GHZ}} \simeq \frac{1}{m} \sum_{k=1}^m \langle S_k \rangle$, where S_k are randomly sampled stabilizers of the GHZ state. Run the circuits on a noisy simulator and, if accessible, on real quantum hardware. Identify the range of n for which genuine multipartite entanglement is achievable, for example by checking whether $F_{\text{GHZ}} > \frac{1}{2}$. Discuss whether a crossover between unitary and dynamic implementations is observed.

2. Hardware-Aware Circuit Design for Long-Range CNOT Gate Teleportation

The efficiency of a dynamic circuit depends strongly on the underlying chip topology.

- 2.1. Choose two distinct qubit topologies—for example, a 2D square, and IBM’s heavy-hexagonal lattice (`ibm_*` backends)—and design long-range CNOT gate teleportation protocols for each. Explain how system qubits and “sacrificial” ancilla bus qubits should be assigned on each topology to maximize effective connectivity.
- 2.2. Implement two variants of a long-range CNOT between two qubits separated by n intermediate qubits:
 - (i) a unitary implementation, either an ancilla-based SWAP strategy from Appendix B of the reference, and the default transpiler output;
 - (ii) a dynamic implementation via mid-circuit measurement and feed-forward.

Report the gate counts, depth, and mid-circuit measurement counts for each.

3. Impact of Noise on the Dynamic Long-Range CNOT

Using the dynamic-circuit implementation from Problem 2, apply different noise models—such as depolarizing noise, readout error, and thermal relaxation (T_1/T_2 noise)—on a simulator and analyze how each noise source degrades the fidelity of the teleported CNOT as n increases. Which noise channel is the dominant limiting factor?

Discuss whether this conclusion is consistent with the error budget

$$\lambda_{\text{tot}} = t_{\text{idle}}\lambda_{\text{idle}} + N_{\text{CNOT}}\lambda_{\text{CNOT}} + N_{\text{meas}}\lambda_{\text{meas}},$$

where each term quantifies the contribution of idle time, CNOT gates, and mid-circuit measurements, respectively, by plugging in the resource counts obtained in Problem 2.

4. Open Questions: Dynamic Circuits Inside Larger Algorithms

The above problems treat long-range CNOT gates and GHZ preparation as isolated primitives. A more practical use of dynamic circuits is to embed them as subroutines inside larger quantum algorithms.

Apply dynamic circuits to quantum algorithms such as the following examples:

- **Example A. Sample-Based Quantum Diagonalization**

Sample-based quantum diagonalization uses bitstrings sampled from a quantum circuit to construct a problem-relevant subspace. Dynamic circuits may be used inside the state-preparation ansatz or in the sampling stage.

- **Example B. Krylov Quantum Diagonalization**

Krylov quantum diagonalization requires estimating overlaps and Hamiltonian matrix elements in a Krylov subspace. These procedures often involve controlled time-evolution operators between an ancilla qubit and a system register.

- **Example C. Variational Quantum Algorithms**

Variational quantum algorithms, including VQE and QAOA, rely on repeated preparation and measurement of parameterized quantum states. Non-local entangling layers in hardware-efficient ansätze often inflate two-qubit depth on restricted-connectivity hardware.

- **Example D. Quantum Phase Estimation**

Quantum phase estimation typically requires controlled powers of a unitary operation and multiple ancilla qubits. Mid-circuit measurement naturally suggests iterative QPE, where a single ancilla qubit is measured, reset, and reused.

Compelling answers to Problem 4 may include the following:

- (i) an explicit small-scale circuit,
- (ii) a resource-count comparison including gate count, depth, and mid-circuit measurement count,
- (iii) a discussion of the regime where dynamic circuits are expected to be advantageous.

References

- [1] E. Bäumer, V. Tripathi, D. S. Wang, P. Rall, E. H. Chen, S. Majumder, A. Seif, and Z. K. Mineev, “Efficient Long-Range Entanglement Using Dynamic Circuits,” *PRX Quantum* **5**, 030339 (2024).
- [2] E. Bäumer, V. Tripathi, A. Seif, D. A. Lidar, and D. S. Wang, “Quantum Fourier Transform Using Dynamic Circuits,” *Physical Review Letters* **133**, 150602 (2024).
- [3] A. D. Córcoles *et al.*, “Exploiting Dynamic Quantum Circuits in a Quantum Algorithm with Superconducting Qubits,” *Physical Review Letters* **127**, 100501 (2021).
- [4] R. Raussendorf, D. E. Browne, and H. J. Briegel, “Measurement-based Quantum Computation on Cluster States,” *Physical Review A* **68**, 022312 (2003).